Biagio Scarciello

Software Engineer

C/C++ | Real-Time Systems | Embedded Linux | Railway Safety-Critical Systems

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Profile

Senior Embedded Software and Systems Engineer with over 8 years of experience in safety-critical railway systems. Strong skills in Embedded C/C++, real-time systems, and communication protocols such as TRDP, IPTCom, and Ethernet. Experienced in Linux environments, system integration, testing, and field deployment.

Work Experience

Embedded Software & Systems Engineer Apr. 2018 – Present

Hitachi Rail STS S.p.A. *Naples, Italy*

- Developed and extended Condition-Based Maintenance (CBM) software using Embedded C, and designed diagnostic algorithms in C and MATLAB/Simulink for onboard systems.
- Acted as System Engineer referent for Bogie Control System communication (TCMS integration) using message data protocol, defining system architecture and coordinating cross-functional validation activities.
- Led integration, troubleshooting, and commissioning of diagnostic subsystems in operational environments, improving reliability of diagnostic data communication between subsystems and TCMS.

High-Speed Platform — *ETR1000 Frecciarossa, ETR 700*

- Led the software development of diagnostic subsystems and coordinated integration, validation, and troubleshooting activities during commissioning.
- Improved reliability and stability of distributed onboard communication networks and managed issue tracking and defect resolution through Jira during deployment phases.

Metro & Mass Transit Platform — *Metro: Rome lines A/B, Milan lines 1, 2, 3; Tram: Firenze, Brescia*

- Technical referent for multiple onboard safety-critical subsystems: Event Recorder systems, Dead Man system, and Zero Speed detection system.
- Led the full software lifecycle of diagnostic systems (from architecture definition to commissioning), developing HMI and optimizing the embedded backend framework in Embedded C.
- Implemented multi-vehicle composition management logic (backend and frontend) and designed and integrated diagnostic HMI components in HTML/JavaScript.

Regional Platform — *Trenitalia Rock, Trenitalia Blues, Ferrovie Nord Milano*

- Successfully delivered and commissioned diagnostic systems for regional train projects.

Key Achievements

- Successfully completed and commissioned the diagnostic systems for the Blues and Rock regional train projects (Trenitalia).
- Enhanced and refactored the embedded diagnostic software framework.
- Designed and delivered the Event Recorder system architecture for the metro lines of Milan, Rome, and Washington (WMATA).
- For the Rome Metro project, led a focused stabilization phase prior to official customer testing, resolving 230 open issues within one month and ensuring successful validation milestones.

Full-Stack Developer Apr. 2024 – Present

Bispace *Naples, Italy*

- Bispace is a personal side project supporting professionals in navigating digital complexity with a clear, human-centered approach.
- **Liftoff** — web application for a professional coach to help clients follow structured training programs and track performance over time, allowing users to access workout plans, monitor progress, and record results. (Python, Flask, HTML, JavaScript)
- **Multicon** — embedded network management application for OpenWRT-based routers (Teltonika RUT955), enabling remote configuration via MQTT JSON commands, with a Flask-based manager server offering a web interface and orchestrating MQTT rooms (topics). (C, Python, Flask, OpenWRT SDK, Mosquitto, JSON, MQTT, WireGuard VPN, Networking)
- **FoxOffice** — web platform to centralize business operations, integrating a CRM module, document management, process automations, and performance dashboards into a single structured environment. (FastAPI, React)
- **NAPP** — multi-tenant HACCP compliance platform for food businesses, built as a monorepo: FastAPI backend, React owner panel for client/document/equipment configuration, and a Flutter tablet app used by operators to log temperatures, production, sanitation, and non-conformities. (FastAPI, PostgreSQL, Alembic, MinIO, React, TypeScript, Flutter, Docker)

- **CRM Evolution Time** — multi-workspace web platform for watch inventory management, with role-based access control (Owner, Admin, Warehouse, Accountant), JWT refresh-token rotation, Excel bulk import, and integration with the OKTOPIA marketplace. (FastAPI, SQLAlchemy 2 async, PostgreSQL, Next.js 15, TypeScript, React Query, Docker)
- **CRM Financial** — accounting management platform for an accounting firm, handling invoices, purchases, daily takings, and consolidated financial summaries, with document storage and multi-client management. (FastAPI, SQLAlchemy async, PostgreSQL, MinIO, React 19, TypeScript, Vite, Docker)
- **Medify** — web application for gynecologists to manage patient records, visits, and pregnancy monitoring.
- **WorkoutLog** — AI-powered web application for tracking and analyzing workouts; users upload sessions via text or image (screenshot), and the system leverages OpenAI APIs to provide personalized feedback and training suggestions based on the user's history and results.
- **Websites** — Sabe SNC, Swinginnapoli, Dott.ssa Federica Terzi, Jtf Gaskets, DV Barberia, Bpnutrition.

Education

M.Sc. in Automation Engineering
University of Naples Federico II | 110/110 cum laude — *Naples, IT*
B.Sc. in Automation Engineering
University of Naples Federico II | 107/110 — *Naples, IT*
Full Stack Development & AI Agents Specialization — In Progress
Start2Impact University (Online)

Technical Skills

Embedded & Real-Time Systems: Embedded C/C++, real-time systems design, memory management & performance optimization, embedded Linux environment, debugging & troubleshooting in field environments
Communication & Distributed Systems: Ethernet-based onboard networks, TRDP (Train Real-time Data Protocol), IPTCom, MQTT, TCP/IP networking, multi-vehicle communication logic, communication gateway design
System Engineering & Validation: System architecture definition, Integration/Verification & Validation (IV&V), commissioning support, requirements analysis & traceability (DOORS), issue tracking (Jira), safety-critical environment exposure (railway domain)
Programming Languages: C (Embedded), C++, Python, JavaScript/TypeScript, Dart
Web & Full-Stack: FastAPI, Flask, React, Next.js, Flutter, SQLAlchemy, PostgreSQL, REST API design, JWT authentication & RBAC, Docker & Docker Compose
Development Tools: CMake, Git, SVN, GitHub, Wireshark, CI/CD fundamentals, MATLAB/Simulink

Languages: *Italian*, *English*
Interests: Crossfit, Swing, Technology, Graphic, Travels

Publications

Model-based vehicular prognostics framework using Big Data architecture, *Computers in Industry, Elsevier*, A. Petrillo, A. Picariello, S. Santini, **B. Scarciello**, G. Sperlì, (2019) - doi.org/10.1016/j.compind.2019.103177

- Proposed **ATOMIC** (Automotive predicTOr Maintenance In Cloud), a cloud-based Big Data infrastructure for Integrated Vehicle Health Management (IVHM) that combines nonlinear discrete-time models of nominal and faulty component behaviour to estimate End-Of-Life (EOL) and Remaining Useful Life (RUL) indicators on-line.
- Validated the prognostic method on the Delphi DFG1596 fuel pump and benchmarked the architecture's deployment configurations in terms of scalability and cost.